Fundamentals of AI and ML Project Report

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YEAR - 1 YEAR
DATE - 31 March 2026

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Problem Statement

In daily life, people often experience symptoms but are unsure about what disease they might indicate. Many individuals rely on random internet searches, which can lead to confusion, misinformation, or panic. There is a lack of simple tools that can help users understand possible health conditions based on symptoms in a structured and reliable way.

This project aims to build a system that takes user symptoms as input and predicts the most likely disease using machine learning, along with recommending basic medicines and precautions.

Why This Problem Matters

This problem is relevant because healthcare awareness is still limited, especially among students and young individuals who may ignore early symptoms. A simple AI-based system can act as a first-level guide and help users make more informed decisions.

The goal is not to replace doctors, but to assist users by providing initial insights and encouraging them to seek proper medical advice when necessary.

Objective

- To build a system that predicts diseases from user symptoms
- To suggest medicines and basic precautions
- To apply machine learning for real-world problem solving
- To create a simple and interactive user interface
- To understand the complete ML workflow from data to prediction

Proposed Solution

- Develop a machine learning-based system that takes user symptoms as input
- Convert symptoms into a structured format for model processing
- Use a Decision Tree model to predict the most likely disease
- Retrieve relevant medicines and precautions from a dataset
- Display results in a clear and user-friendly format
- Provide an interactive system that allows users to recheck symptoms easily

Approach and Methodology

Data Collection

>Two datasets were used:

- Disease–symptom dataset (from Kaggle)
- Medicine and precautions dataset

Data Preprocessing

- Cleaned inconsistent column names
- Handled missing and incorrect valuess
- Converted symptom data into binary format (0/1)

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- Handled missing and incorrect values
- Converted symptom data into binary format (0/1)
- Ensured uniform formatting for disease names

Feature Engineering

>Symptoms were converted into binary feature vectors where:

- 1 → symptom present
- 0 → symptom absent

>This allowed the machine learning model to process the data effectively.

Model Selection

A Decision Tree Classifier was used because:

- It is simple and interpretable
- Works well with categorical/binary data
- Suitable for small-to-medium datasets

Prediction System

- User inputs symptoms
- Model predicts disease
- System compares prediction confidence

Key Features

- Disease prediction based on symptoms
- Medicine recommendation system
- Confidence comparison of diseases
- Interactive command-line interface
- User input (name and age)
- Reusable system for multiple checks
- Structured diagnosis report output

Key Decisions Made

1. Choosing a Real-Life Problem

- Focused on healthcare for practical impact

2. Using Decision Tree Model

- Selected for simplicity and efficiency

3. Keeping Interface Simple

- Used CLI instead of GUI for easy execution

4. Flexible Matching Logic

- Ensured correct mapping between disease and medicines

Challenges Faced

- Dataset inconsistencies and formatting issues
- CSV parsing errors
- Converting text symptoms into numerical format
- Matching predicted diseases with medicine dataset
- Handling user input and improving usability

Results and Observations

- The system successfully predicts diseases based on user-input symptoms
- The model is able to identify patterns between symptoms and diseases
- The system works smoothly in an interactive command-line environment
- Results improve with better quality and structured data

- It provides relevant medicine suggestions and basic precautions
- Confidence levels help compare multiple possible diseases
- The overall system demonstrates effective application of machine learning in a real-world scenario

Limitations

- Limited dataset size
- Predictions may not always be accurate
- No graphical interface
- Not a replacement for professional medical advice

What I Learned

Technical Skills

- Python programming
- Data preprocessing
- Machine learning model training
- Error handling and debugging

Conceptual Understanding

- Importance of clean data

- Feature engineering
- Real-world application of AI/ML

Practical Insights

- Simple models can solve real problems
- Data quality affects performance
- AI systems improve over time

Conclusion

The Disease Prediction and Medicine Recommendation System successfully addresses the problem of identifying diseases from symptoms. It combines machine learning with a simple interface to create a practical and useful solution.

This project demonstrates how AI/ML concepts can be applied to real-world healthcare problems in an effective and understandable way.

References

- Python Documentation
- Scikit-learn Documentation
- Course Materials (AI & ML)